DO NOW — RECALL

Programming Techniques · Scope and Lifetime

Name: _____ Class: _____ Date: _____

1. What does return do in a method?

2. What is a condition-controlled loop?

3. Why might binary search outperform linear search?

4. Simplify ¬¬B.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Sends a value back to its caller

2. Model answer

Repeats while a test holds

3. Model answer

It halves the remaining range

4. Model answer

B

By the end, you can…

- Distinguish local and class-level variables
- Trace shadowed identifiers
- Avoid unintended shared state

Vocabulary — know these terms

Recall the term — scope

Recall the term — local

Recall the term — class field

Recall the term — shadowing

Recall the term — lifetime

NEW LEARNING

Read, annotate and explain

Spot two variables with one name

class Program { static int score = 10; static void Main() { int score = 3; System.Console.WriteLine(score); System.Console.WriteLine(Program.score); } }

Explain the important decision

The class field and method-local variable are different storage locations. C# allows a local to hide a field; this does not permit redeclaring any local in a nested block.

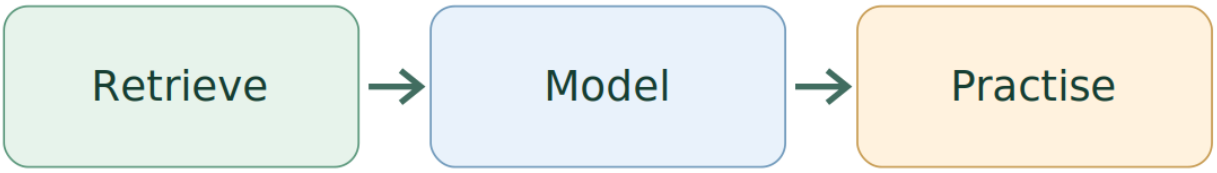
Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Spot two variables with one name

class Program { static int score = 10; static void Main() { int score = 3; System.Console.WriteLine(score); System.Console.WriteLine(Program.score); } }



Three-stage diagram showing retrieve, model, and practise

I DO — WORKED EXAMPLE

Follow the worked method

Spot two variables with one name

Identify both variables

Program.score is a class-level static field, initially 10. Main declares its own local score, initially 3.

Resolve each name

The unqualified score inside Main refers to the local. Program.score explicitly selects the class field; the field has not changed.

Explain the output and scope

The output is 3 then 10. Another method cannot directly access Main's local score. Restricting scope reduces unintended changes to shared state.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

1. Problem 1

A parameter n is set to 4. What is visible inside the function?

Method / working:

2. Problem 2

Why avoid excessive global state?

Method / working:

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Problem 1 A parameter n is set to 4. What is visible inside the function? Discuss, then respond

DISCUSS — Compare approaches, walk through one test input, and explain how you would detect or correct a mistake.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

Guided 2: Problem 2 Why avoid excessive global state?

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Apply independently
Explain one benefit of local variables. [2]

Question 2

Apply independently
A method declares int total=0. Can main read total directly? [1]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Apply independently
Define variable shadowing. [2]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

True or false: every local variable is visible throughout the program.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Check your thinking

The parameter n with value 4.

IMPROVE: Explain one correction to your own reasoning.

Guided 2 — original CHECK

Discuss, then respond

Why avoid excessive global state?

Check your thinking

Changes from one part of the program can affect another.

IMPROVE: Explain one correction to your own reasoning.